Module 04 Assignments

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October 17, 2026

Group 8.1

Question 2

Use the arc length formula to find the length of the curve $y = \sqrt{4 - x^2}$, $0 \leq x \leq 2$. Check your answer by noting that the curve is part of a circle.

Differentiate using the chain rule

$$\frac{dy}{dx} = -\frac{x}{\sqrt{4 - x^2}}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \frac{x^2}{4 - x^2}$$

Set up the arc length integral

$$L = \int_0^2 \sqrt{1 + \frac{x^2}{4 - x^2}}\, dx$$

Combine fractions and simplify

$$L = \int_0^2 \sqrt{\frac{4}{4 - x^2}}\, dx$$

$$L = \int_0^2 \frac{2}{\sqrt{4 - x^2}}\, dx$$

Evaluate the definite integral

$$L = \left[2 \arcsin\left(\frac{x}{2}\right)\right]_0^2$$

$$= (\pi) - (0)$$

$$\boxed{\pi}$$

Verify using the geometry of a circle

$$y = \sqrt{4 - x^2}$$

This is recognized as a circle by squaring both sides

$$x^2 + y^2 = 4$$

The standard equation of a circle is

$$x^2 + y^2 = r^2$$

Therefore

$$r^2 = 4 \Longrightarrow r = 2$$

Using the circumference formula

$$L = \frac{1}{4}(2\pi r) = \frac{1}{4}(2\pi \cdot 2)$$

$$\boxed{L = \pi}$$

Question 3

Set up, but do not evaluate, an integral for the length of the curve.

$$y = x^3, \qquad 0 \leq x \leq 2$$

Differentiate using the chain rule

$$\frac{dy}{dx} = 3x^2$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = 9x^4$$

Set up the arc length integral

$$\boxed{L = \int_0^2 \sqrt{1 + 9x^4}\, dx}$$

Question 4

Set up, but do not evaluate, an integral for the length of the curve.

$$y = e^x, \qquad 1 \leq x \leq 3$$

Differentiate

$$\frac{dy}{dx} = e^x$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = e^{2x}$$

Set up the arc length integral

$$\boxed{L = \int_1^3 \sqrt{1 + e^{2x}}\, dx}$$

Question 7

Set up, but do not evaluate, an integral for the length of the curve.

$$x = \sin(y), \qquad 0 \leq y \leq \frac{\pi}{2}$$

Differentiate

$$\frac{dx}{dy} = \cos(y)$$

Square the derivative

$$\left(\frac{dx}{dy}\right)^2 = \cos^2(y)$$

Set up the arc length integral

$$\boxed{L = \int_0^{\frac{\pi}{2}} \sqrt{1 + \cos^2(y)}\, dy}$$

Question 8

Set up, but do not evaluate, an integral for the length of the curve.

$$y^2 = \ln x, \qquad -1 \leq y \leq 1$$

Exponentiate both sides

$$e^{y^2} = e^{\ln(x)}$$

$$e^{y^2} = x$$

Differentiate with respect to y

$$\frac{dx}{dy} = 2ye^{y^2}$$

Square the derivative

$$\left(\frac{dx}{dy}\right)^2 = 4y^2e^{2y^2}$$

Set up the arc length integral

$$\boxed{L = \int_{-1}^{1} \sqrt{1 + 4y^2e^{2y^2}}\, dy}$$

Question 9

Find the exact length of the curve.

$$y = \frac{2}{3}x^{\frac{3}{2}}, \qquad 0 \leq x \leq 2$$

Differentiate using the power rule

$$\frac{dy}{dx} = \sqrt{x}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = x$$

Set up the arc length integral

$$L = \int_0^2 \sqrt{1+x}\, dx$$

Apply u-substitution

Let

$$u = 1 + x \qquad du = dx$$

Substitute and integrate

$$L = \int u^{\frac{1}{2}}\, du$$

$$= \left(\frac{2}{3}u^{\frac{3}{2}}\right)$$

Evaluate the bounds

$$L = \left[\frac{2}{3}(1+x)^{\frac{3}{2}}\right]_0^2$$

$$= \frac{2}{3}(3\sqrt{3}) - \left(\frac{2}{3}\right)$$

$$\boxed{2\sqrt{3} - \frac{2}{3}}$$

Question 11

Find the exact length of the curve.

$$y = \frac{2}{3}(1+x^2)^{\frac{3}{2}}, \qquad 0 \le x \le 1$$

Differentiate using the chain rule

$$\frac{dy}{dx} = \frac{2}{3} \cdot \frac{3}{2}(1+x^2)^{\frac{1}{2}} \cdot 2x$$

$$= 2x(\sqrt{1+x^2})$$

Square the derivative

$$\left(2x(\sqrt{1+x^2}\right)^2 = 4x^2(1+x^2)$$

Set up the arc length integral

$$L = \int_0^1 \sqrt{1+4x^2+4x^4}\,dx$$

Simplify

$$L = \int_0^1 \sqrt{(2x^2+1)^2}\,dx$$

Integrate using the power rule

$$L = \left[\frac{2}{3}x^3 + x\right]_0^1$$

Evaluate at the bounds

$$= \left(\frac{5}{3}\right) - (0)$$

$$\boxed{\frac{5}{3}}$$

Question 13

Find the exact length of the curve.

$$y = \frac{x^3}{3} + \frac{1}{4x}, \qquad 1 \le x \le 2$$

Differentiate using the power rule

$$\frac{dy}{dx} = x^2 - \frac{1}{4x^2}$$

Square the derivative

$$\left(x^2 - \frac{1}{4x^2}\right)^2 = x^4 - \frac{1}{2} + \frac{1}{16x^4}$$

Set up the arc length integral

$$L = \int_1^2 \sqrt{1 + x^4 - \frac{1}{2} + \frac{1}{16x^4}}\, dx$$

Combine like terms

$$L = \int_1^2 \sqrt{x^4 + \frac{1}{2} + \frac{1}{16x^4}}\, dx$$

Factor the perfect square trinomial

$$= \int_1^2 \sqrt{\left(x^2 + \frac{1}{4x^2}\right)^2}\, dx$$

Simplify the radical

$$L = \int_1^2 x^2 + \frac{1}{4x^2}\, dx$$

Integrate using the power rule

$$L = \left[\frac{x^3}{3} - \frac{1}{4x}\right]_1^2$$

Evaluate at the bounds

$$= \left(\frac{61}{24}\right) - \left(\frac{1}{12}\right)$$

$$\boxed{\frac{59}{24}}$$

Question 16

Find the exact length of the curve.

$$y = \ln(\cos(x)), \qquad 0 \le x \le \frac{\pi}{3}$$

Differentiate using the chain rule

$$\frac{dy}{dx} = -\frac{\sin(x)}{\cos(x)}$$

Recall the trigonometric identity

$$\frac{\sin(x)}{\cos(x)} = \tan(x)$$

Therefore

$$\frac{dy}{dx} = -\tan(x)$$

Square the derivative

$$(-\tan(x))^2 = \tan^2(x)$$

Set up the arc length integral

$$L = \int_0^{\frac{\pi}{3}} \sqrt{1 + \tan^2(x)}\, dx$$

Apply the Pythagorean identity

$$1 + \tan^2(x) = \sec^2(x)$$

Since $sec^2(x)$ is positive on our interval $0 \leq x \leq \frac{\pi}{3}$, we have

$$\sqrt{\sec^2(x)} = \sec(x)$$

$$L = \int_0^{\frac{\pi}{3}} \sec(x)\, dx$$

Integrate using the secant formula

$$\int_0^{\frac{\pi}{3}} \sec(x)\, dx = \ln|\sec(x) + \tan(x)|$$

Evaluate the bounds

$$[\ln|\sec(x) + \tan(x)|]_0^{\frac{\pi}{3}}$$

Upper bound $x = \frac{\pi}{3}$

$$F\left(\frac{\pi}{3}\right) = \ln\left|\sec\left(\frac{\pi}{3}\right) + \tan\left(\frac{\pi}{3}\right)\right|$$

$$= \ln(2 + \sqrt{3})$$

Lower bound $x = 0$

$$F\left(0\right) = \ln\left|\sec(0) + \tan(0)\right|$$

$$= \ln\left|1 + 0\right|$$

$$= 0$$

Subtract

$$= \ln(2 + \sqrt{3}) - 0$$

$$\boxed{\ln(2 + \sqrt{3})}$$

Group 8.2

Question 1

The given curve is rotated about the x-axis. Set up, but do not evaluate, an integral for the area of the resulting surface by integrating

$$y = \sqrt[3]{x}, \qquad 1 \leq x \leq 8$$

a. with respect to x

Differentiate

$$\frac{dy}{dx} = \frac{1}{3}(x)^{-\frac{2}{3}}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left[\frac{1}{3}(x)^{-\frac{2}{3}}\right]^2$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{1}{9}(x)^{-\frac{4}{3}}$$

Set up the surface area integral

For rotation about the x-axis

$$S = 2\pi \int_a^b y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Substitute the function, squared derivative, and bounds

$$\boxed{S = \int_1^8 2\pi \sqrt[3]{x}\sqrt{1 + \frac{1}{9}(x)^{-\frac{4}{3}}}\, dx}$$

b. with respect to y.

Solve for x

$$y = (x)^{\frac{1}{3}}$$

Cube both sides

$$y^3 = x$$

Differentiate with respect to y

$$\frac{dx}{dy} = \frac{d}{dy}\left(3y^2\right)$$

Square the derivative

$$\left(\frac{dx}{dy}\right)^2 = \left(3y^2\right)^2$$

$$\left(\frac{dx}{dy}\right)^2 = 9y^4$$

Convert the integration bounds

$$y = \sqrt[3]{x}$$

Lower bound, $x = 1$

$$y = \sqrt[3]{1} = 1$$

Upper bound, $x = 8$

$$y = \sqrt[3]{8} = 2$$

Therefore

$$1 \leq y \leq 2$$

Set upe the surface area integral

For rotation about the x-axis, integrating with respect to y

$$S = 2\pi \int_c^d y\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

Substitute the squared derivative and converted bounds

$$\boxed{S = \int_1^2 2\pi y\sqrt{1 + 9y^4}\, dy}$$

Question 3

The given curve is rotated about the x-axis. Set up, but do not evaluate, an integral for the area of the resulting surface by integrating

$$x = \ln(2y + 1), \qquad 0 \leq y \leq 1$$

a. with respect to x and

Solve for y

$$x = \ln(2y + 1)$$

Exponentiate both sides to eliminate the logarithm

$$e^x = e^{\ln(2y+1)} = 2y + 1$$

Subtract 1, then divide both sides by 2

$$y = \frac{1}{2}(e^x - 1)$$

Differentiate with respect to x

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{2}e^x\right)$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{1}{2}e^x\right)^2$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{1}{4}e^{2x}$$

Convert the integration bounds

$$x = \ln(2y + 1)$$

Lower bound, $y = 0$

$$x = \ln(2(0) + 1) = 0$$

Upper bound, $y = 1$

$$x = \ln(2(1) + 1) = \ln 3$$

Therefore

$$0 \le x \le \ln 3$$

Set up the surface area integral

$$S = \int_0^{\ln 3} 2\pi \frac{1}{2}(e^x - 1)\sqrt{1 + \frac{1}{4}e^{2x}}\, dx$$

Simplify the coefficient

$$\boxed{S = \int_0^{\ln 3} \pi(e^x - 1)\sqrt{1 + \frac{1}{4}e^{2x}}\, dx}$$

b. with respect to y.

Differentiate with respect to y

$$\frac{d}{dy}\left[\ln(2y+1)\right] = \frac{2}{2y+1}$$

Square the derivative

$$\left(\frac{dx}{dy}\right)^2 = \left(\frac{2}{2y+1}\right)^2$$

$$\left(\frac{dx}{dy}\right)^2 = \frac{4}{(2y+1)^2}$$

Set up the surface area integral

$$\boxed{S = \int_0^1 2\pi y\sqrt{1 + \frac{4}{(2y+1)^2}}\, dy}$$

Question 6

The given curve is rotated about the y-axis. Set up, but do not evaluate, an integral for the area of the resulting surface by integrating

$$y = (x+1)^4, \qquad 0 \le x \le 2$$

a. with respect to x

Differentiate with respect to x

$$\frac{dy}{dx} = 4(x+1)^3$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left[4(x+1)^3\right]^2$$

$$\left(\frac{dy}{dx}\right)^2 = 16(x+1)^6$$

Set up the surface area integral

For rotation about the y-axis, integrating with respect to x

$$\boxed{S = \int_0^2 2\pi x\sqrt{1+16(x+1)^6}\,dx}$$

b. with respect to y.

First solve for x.

$$x = \sqrt[4]{y} - 1$$

Differentiate with respect to y

$$\frac{dx}{dy} = \frac{d}{dy}y^{\frac{1}{4}} - 1$$

Apply the power rule

$$\frac{dx}{dy} = \frac{1}{4}y^{-\frac{3}{4}}$$

Square the derivative

$$\left(\frac{dx}{dy}\right)^2 = \left(\frac{1}{4}y^{-\frac{3}{4}}\right)^2$$

$$\left(\frac{dx}{dy}\right)^2 = \frac{1}{16}y^{-\frac{3}{2}}$$

Convert the bounds

Use the original function

$$y = (x+1)^4$$

Lower bound, $x = 0$

$$y = (0+1)^4$$

$$= 1$$

Upper bound, $x = 2$

$$y = (2+1)^4$$

$$= 81$$

Therefore

$$1 \leq y \leq 81$$

Set up the surface area integral

For rotation about the y-axis, integrating with respect to y

$$\boxed{S = \int_1^{81} 2\pi(\sqrt[4]{y} - 1)\sqrt{1 + \frac{1}{16}y^{-\frac{3}{2}}}\, dy}$$

Question 7

The given curve is rotated about the y-axis. Set up, but do not evaluate, an integral for the area of the resulting surface by integrating

$$y = 1 + \sin(x), \qquad 0 \leq x \leq \frac{\pi}{2}$$

a. with respect to x

Differentiate

$$\frac{dy}{dx} = \cos(x)$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = (\cos(x))^2$$

$$\left(\frac{dy}{dx}\right)^2 = \cos^2(x)$$

Set up the surface area integral

For rotation about the y-axis with respect to x

$$\boxed{S = \int_0^{\frac{\pi}{2}} 2\pi x\sqrt{1+\cos^2(x)}\,dx}$$

b. with respect to y.

Solve for x

$$y = 1 + \sin(x)$$

Isolate the sine function

$$y - 1 = \sin(x)$$

Apply the inverse sine

$$x = \arcsin(y-1)$$

Differentiate with respect to y

$$\frac{dx}{dy} = \frac{d}{dy}(\arcsin(y-1))$$

Apply the chain rule

Let

$$u = y - 1 \qquad \frac{du}{dy} = 1$$

Therefore

$$\frac{d}{dy}(\arcsin(y-1)) = \frac{1}{\sqrt{1-(y-1)^2}}$$

Square the derivative

$$\left(\frac{dx}{dy}\right)^2 = \left(\frac{1}{\sqrt{1-(y-1)^2}}\right)^2$$

$$\left(\frac{dx}{dy}\right)^2 = \frac{1}{1-(y-1)^2}$$

Convert the integration bounds

Use the original function

$$y = 1 + \sin(x)$$

Lower bound, $x = 0$

$$y = 1 + \sin(0)$$

$$= 0$$

Upper bound, $x = \frac{\pi}{2}$

$$y = 1 + \sin(\frac{\pi}{2})$$

$$= 1$$

Set up the surface area integral

For rotation about the y-axis, integrating with respect to y

$$S = \int_0^1 2\pi \arcsin(y-1)\sqrt{1 + \frac{1}{1-(y-1)^2}}\, dy$$

Simplify the denominator

$$\boxed{S = \int_0^1 2\pi \arcsin(y-1)\sqrt{1 + \frac{1}{2y-y^2}}\, dy}$$

Question 10

Find the exact area of the surface obtained by rotating the curve about the x-axis.

$$y = \sqrt{5-x}, \qquad 3 \leq x \leq 5$$

Differentiate using the chain rule

$$y = (5-x)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = -\frac{1}{2}(5-x)^{-\frac{1}{2}}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left[-\frac{1}{2}(5-x)^{-\frac{1}{2}}\right]^2$$

$$= \frac{1}{4}(5-x)^{-1}$$

Set up the surface integral

$$S = \int_3^5 2\pi\sqrt{5-x}\sqrt{1+\frac{1}{4}(5-x)^{-1}}\,dx$$

Simplify the rational expression

$$1+\frac{1}{4}(5-x)^{-1} = 1+\frac{1}{4(5-x)}$$

$$= 1+\frac{1}{20-4x}$$

$$= \frac{20-4x}{20-4x}+\frac{1}{20-4x}$$

$$= \frac{21-4x}{20-4x}$$

Simplify the integrand

$$S = \int_3^5 2\pi\sqrt{5-x}\sqrt{\frac{21-4x}{20-4x}}\,dx$$

Factor the denominator

$$S = \int_3^5 2\pi\sqrt{5-x}\sqrt{\frac{21-4x}{4(5-x)}}\,dx$$

Combine the radicals and cancel the common factor

$$\frac{\sqrt{21-4x}}{\sqrt{4}} = \frac{1}{2}\sqrt{21-4x}$$

Therefore

$$S = \int_3^5 2\pi \frac{1}{2}\sqrt{21-4x}\,dx$$

$$= \int_3^5 \pi\sqrt{21-4x}\,dx$$

Apply u-substitution

Let

$$u = 21-4x \qquad du = -4\,dx \qquad dx = -\frac{1}{4}\,du$$

Convert the integration bounds

Lower bound, $x = 3$

$$u = 21-4(3) = 9$$

Upper bound, $x = 5$

$$u = 21-4(5) = 1$$

Substitute and reverse the bounds to remove the negative sign

$$S = -\frac{\pi}{4}\int_9^1 u^{\frac{1}{2}}\,dx$$

$$S = \frac{\pi}{4}\int_1^9 u^{\frac{1}{2}}\,dx$$

Find the antiderivative

$$S = \frac{\pi}{4}\left(\frac{2}{3}u^{\frac{3}{2}}\right)$$

$$S = \frac{\pi}{6}\left[u^{\frac{3}{2}}\right]_1^9$$

Evaluate the bounds

$$S = \frac{27\pi}{6} - \frac{\pi}{6}$$

$$\boxed{\frac{13\pi}{3}}$$

Question 11

Find the exact area of the surface obtained by rotating the curve about the x-axis.

$$y^2 = x + 1, \qquad 0 \le x \le 3$$

Differentiate using the chain rule

$$y = (x+1)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(x+1)^{-\frac{1}{2}}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left[-\frac{1}{2}(x+1)^{-\frac{1}{2}}\right]^2$$

$$= \frac{1}{4}(x+1)^{-1}$$

Set up the surface integral

$$S = \int_0^3 2\pi\sqrt{x+1}\sqrt{1 + \frac{1}{4}(x+1)^{-1}}\,dx$$

Simplify the rational expression

$$1 + \frac{1}{4}(x+1)^{-1} = 1 + \frac{1}{4(x+1)}$$

$$= 1 + \frac{1}{4x+4}$$

$$= \frac{4x+4}{4x+4x} + \frac{1}{4x+4}$$

$$= \frac{4x+5}{4x+4}$$

Simplify the integrand

$$S = \int_0^3 2\pi\sqrt{x+1}\sqrt{\frac{4x+5}{4x+4}}\,dx$$

Factor the denominator

$$S = \int_0^3 2\pi\sqrt{x+1}\sqrt{\frac{4x+5}{4(x+1)}}\,dx$$

Combine the radicals and cancel the common factor

$$\frac{\sqrt{4x+5}}{\sqrt{4}} = \frac{1}{2}\sqrt{4x+5}$$

Therefore

$$S = \int_0^3 2\pi\frac{1}{2}\sqrt{4x+5}\,dx$$

$$= \int_0^3 \pi\sqrt{4x+5}\,dx$$

Apply u-substitution

Let

$$u = 4x+5 \qquad du = 4\,dx \qquad dx = \frac{1}{4}\,du$$

Convert the integration bounds

Lower bound, $x = 0$

$$u = 4(0)+5 = 5$$

Upper bound, $x = 3$

$$u = 4(3)+5 = 17$$

Substitute the bounds

$$S = \frac{\pi}{4}\int_5^{17} u^{\frac{1}{2}}\,dx$$

Find the antiderivative

$$S = \frac{\pi}{4}\left(\frac{2}{3}u^{\frac{3}{2}}\right)$$

$$S = \frac{\pi}{6}\left[u^{\frac{3}{2}}\right]_5^{17}$$

Evaluate the bounds

$$S = \frac{\pi 17\sqrt{17} - \pi 5\sqrt{5}}{6}$$

$$\boxed{\frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})}$$

Question 12

Find the exact area of the surface obtained by rotating the curve about the x-axis.

$$y = \sqrt{1+e^x}, \qquad 0 \le x \le 1$$

Differentiate using the chain rule

$$y = (1+e^x)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{e^x}{2}(1+e^x)^{-\frac{1}{2}}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left[\frac{e^x}{2}(1+e^x)^{-\frac{1}{2}}\right]^2$$

$$= \frac{e^{2x}}{4}(1+e^x)^{-1}$$

Set up the surface integral

For rotation about the x-axis

$$S = \int_0^1 2\pi\sqrt{1+e^x}\sqrt{1+\frac{e^{2x}}{4}(1+e^x)^{-1}}\,dx$$

Simplify the rational expression

$$1+\frac{e^{2x}}{4}(1+e^x)^{-1} = 1+\frac{e^{2x}}{4(1+e^x)}$$

$$= 1+\frac{e^{2x}}{4+4e^x}$$

$$= \frac{4+4e^x}{4+4e^x}+\frac{e^{2x}}{4+4e^x}$$

$$= \frac{4+4e^x+e^{2x}}{4+4e^x}$$

Simplify the integrand

$$S = \int_0^1 2\pi\sqrt{1+e^x}\sqrt{\frac{4+4e^x+e^{2x}}{4+4e^x}}\,dx$$

Factor the denominator and perfect-square trinomial

$$S = \int_0^1 2\pi\sqrt{1+e^x}\sqrt{\frac{(2+e^x)^2}{4(1+e^x)}}\,dx$$

Combine the radicals and cancel the common factor

$$\sqrt{\frac{(2+e^x)^2}{4}} = \frac{2+e^x}{2}$$

Therefore

$$S = \int_0^1 2\pi\frac{2+e^x}{2}\,dx$$

$$= \int_0^1 \pi(2+e^x)\,dx$$

Find the antiderivative

$$= 2x+e^x$$

$$S = \pi \left[2x + e^x\right]_0^1$$

Evaluate the bounds

Upper bound $x = 1$

$$\pi(2(1) + e^1) = \pi(2 + e^x)$$

Lower bound $x = 0$

$$\pi(2(0) + e^0) = \pi$$

Subtract

$$S = \pi(2 + e^x) - \pi$$

$$\pi(2 + e^x - 1)$$

$$\boxed{\pi(1 + e^x)}$$

Question 15

Find the exact area of the surface obtained by rotating the curve about the x-axis.

$$x = \frac{1}{3}(y^2 + 2)^{\frac{3}{2}}, \qquad 1 \leq y \leq 2$$

Differentiate using the chain rule

$$\frac{dx}{dy} = \frac{1}{3} \cdot \frac{3}{2} \cdot 2y(y^2 + 2)^{\frac{1}{2}}$$

$$\frac{dx}{dy} = y(y^2 + 2)^{\frac{1}{2}}$$

Square the derivative

$$\left(\frac{dx}{dy}\right)^2 = \left[y(y^2 + 2)^{\frac{1}{2}}\right]^2$$

$$= y^2(y^2 + 2)$$

Expand the expression

$$= y^4 + 2y^2$$

Set up the surface integral

For rotation about the x-axis

$$S = 2\pi \int_1^2 y\sqrt{1 + y^4 + 2y^2}\, dy$$

Factor the perfect square trinomial

$$S = 2\pi \int_1^2 y\sqrt{(y^2 + 1)^2}\, dy$$

$$2\pi \int_1^2 (y^3 + y)\, dy$$

Find the antiderivative

$$= 2\pi \left[\frac{y^4}{4} + \frac{y^2}{2}\right]_1^2$$

Evaluate the bounds

Apply the Fundamental Theorem of Calculus

$$S = 2\pi \left[\left(\frac{2^4}{4} + \frac{2^2}{2}\right) - \left(\frac{1^4}{4} + \frac{1^2}{2}\right)\right]$$

Evaluate the upper bound

$$\frac{2^4}{4} + \frac{2^2}{2} = \frac{16}{4} + \frac{4}{2}$$

$$= 4 + 2$$

$$= 6$$

Evaluate the lower bound

$$\frac{1^4}{4} + \frac{1^2}{2} = \frac{1}{4} + \frac{1}{2}$$

$$= \frac{3}{4}$$

Subtract and simplify

$$S = 2\pi\left(6 - \frac{3}{4}\right)$$

$$= 2\pi\left(\frac{24}{4} - \frac{3}{4}\right)$$

$$\boxed{\frac{21\pi}{2}}$$

Question 16

Find the exact area of the surface obtained by rotating the curve about the x-axis.

$$x = 1 + 2y^2, \qquad 1 \leq y \leq 2$$

Differentiate using the power rule

$$\frac{dx}{dy} = 4y$$

Square the derivative

$$\left(\frac{dx}{dy}\right) = (4y)^2$$

$$= 16y^2$$

Set up the surface area integral

$$S = 2\pi \int_1^2 y\sqrt{1 + 16y^2}\, dy$$

Let

$$u = 1 + 16y^2 \qquad du = 32y\, dy \qquad y\, dy = \frac{1}{32}\, du$$

Convert the integration bounds

Lower bound, $y = 1$

$$u = 1 + 16(1)^2 = 17$$

Upper bounds, $y = 2$

$$u = 1 + 16(2)^2 = 65$$

Rewrite the integral in terms of u

$$S = 2\pi \int_{17}^{65} u^{\frac{1}{2}} \left(\frac{1}{32}\right) du$$

$$= \frac{\pi}{16} \int_{17}^{65} u^{\frac{1}{2}} \, du$$

Find the derivative

$$S = \frac{\pi}{16} \left[\frac{2}{3} u^{\frac{3}{2}}\right]_{17}^{65}$$

$$S = \frac{\pi}{24} \left[u^{\frac{3}{2}}\right]_{17}^{65}$$

Evaluate the bounds

$$S = \frac{\pi}{24} \left(65^{\frac{3}{2}} - 17^{\frac{3}{2}}\right)$$

$$\boxed{\frac{\pi}{24} \left(65\sqrt{65} - 17\sqrt{17})\right)}$$

Question 17

The given curve is rotated about the y-axis. Find the area of the resulting surface.

$$y = \frac{1}{3} x^{\frac{3}{2}}, \qquad 0 \leq x \leq 12$$

with respect to x

Differentiate

$$\frac{dy}{dx} = \frac{1}{3} \cdot \frac{3}{2} x^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2} x^{\frac{1}{2}}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{1}{2}x^{\frac{1}{2}}\right)^2$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{x}{4}$$

Set up the surface area integral

For rotation about the y-axis with respect to x

$$S = 2\pi \int_0^{12} x\sqrt{1+\frac{x}{4}}\,dx$$

Let

$$u = 1 + \frac{x}{4} \qquad du = \frac{1}{4}\,dx \qquad dx = 4\,du$$

Express x in terms of u

$$u - 1 = \frac{x}{4}$$

$$x = 4(u-1)$$

Convert the integration bounds

Lower bound, $y = 0$

$$u = 1 + \frac{0}{4} = 1$$

Upper bounds, $y = 12$

$$u = 1 + \frac{12}{4} = 4$$

Rewrite the integral in terms of u

$$S = 2\pi \int_1^4 4(u-1)u^{\frac{1}{2}}\,(4)\,du$$

$$S = 32\pi \int_1^4 u^{\frac{3}{2}} - u^{\frac{1}{2}}\,du$$

Find the derivative

$$S = 32\pi \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_1^4$$

Evaluate the bounds

$$S = 32\pi \left[\left(\frac{2}{5} \cdot 4^{\frac{5}{2}} - \frac{2}{3} \cdot 4^{\frac{3}{2}} \right) - \left(\frac{2}{5} \cdot 1^{\frac{5}{2}} - \frac{2}{3} \cdot 1^{\frac{3}{2}} \right) \right]$$

Upper bound

$$\frac{2}{5} \cdot 4^{\frac{5}{2}} - \frac{2}{3} \cdot 4^{\frac{3}{2}}$$

$$= \frac{2}{5}(32) - \frac{2}{3}(8)$$

$$= \frac{64}{5} - \frac{16}{3}$$

$$= \frac{112}{15}$$

Lower bound

$$\frac{2}{5} \cdot 1^{\frac{5}{2}} - \frac{2}{3} \cdot 1^{\frac{3}{2}}$$

$$= \frac{2}{5} - \frac{2}{3}$$

$$= -\frac{4}{15}$$

$$32\pi \left[\frac{112}{15} - \left(-\frac{4}{15} \right) \right]$$

$$32\pi \left(\frac{116}{15} \right)$$

$$\boxed{\frac{3712\pi}{15}}$$

Question 33

The surface formed by rotating the curve $y = \frac{1}{x}$, $x \geq 1$, about the x-axis is known as Gabriel's horn. Show that the surface area is infinite (although the enclosed volume is finite; see Exercise 7.8.75).

Show that the integrand is greater than $\frac{1}{x}$ and integrate $\frac{1}{x}\,dx$ from one to infinity.

Surface area formula

For rotation about the x-axis

$$S = 2\pi \int_a^b y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$$

Rewrite using a negative exponent

$$y = \frac{1}{x} = x^{-1}$$

Differentiate using the power rule

$$y = x^{-1} = -x^{-2}$$

Square the derivative

$$\left(\frac{dy}{dx}\right)^2 = \left(-x^{-2}\right)^2$$

$$= x^{-4}$$

Set up the improper integral

$$S = 2\pi \int_1^\infty \frac{1}{x}\sqrt{1 + x^{-4}}\,dx$$

Rewrite as a limit

$$S = 2\pi \lim_{t\to\infty} \int_1^t \frac{1}{x}\sqrt{1 + x^{-4}}\,dx$$

Establish the comparison inequality

Since $x \geq 2$

$$x^{-4} > 0$$

$$1 + x^{-4} > 1$$

$$\sqrt{1 + x^{-4}} > 1$$

Since $\frac{1}{x} > 0$, multiply both sides by $\frac{1}{x}$ without reversing the inequality

$$\frac{1}{x}\sqrt{1 + x^{-4}} > \frac{1}{x}$$

The surface area integrand is greater than the simpler function $\frac{1}{x}$

Evaluate the comparison integral

$$\int_1^\infty \frac{1}{x}\,dx$$

Rewrite as a limit

$$\lim_{t\to\infty} \int_1^t \frac{1}{x}\,dx$$

Use the logarithm integration rule

$$\lim_{t\to\infty} [\ln x]_1^t$$

$$\lim_{t\to\infty} (\ln t - 0)$$

As $t \to \infty$

$$\ln t \to \infty$$

Thus

$$\int_1^t \frac{1}{x}\,dx \quad \text{diverges}$$

Apply the Direct Comparison Test

$$0 < \frac{1}{x} < \frac{1}{x}\sqrt{1 + x^{-4}} \qquad (x \geq 1)$$

$$\int_1^\infty \frac{1}{x}\,dx \quad \text{diverges}$$

By the Direct Comparison Test, the integral of the larger, nonnegative function must also diverge

$$\int_1^\infty \frac{1}{x}\sqrt{1+x^{-1}}\,dx = \infty$$

$$\boxed{S = \infty}$$
